AI PRODUCT MANAGEMENT CASE STUDY

ServiceTrust AI

AI-Powered Non-Human Identity Governance

From 5,000 unmanaged service accounts to autonomous risk detection & remediation for Insurance under NYDFS 23 NYCRR 500

Product: Autonomous NHI Governance Platform | Owner: IAM + SOC | Sponsors: CISO & CTO

Stack: Azure Entra ID (99%) + SailPoint + CyberArk + ServiceNow + Sentinel + ADX + Azure OpenAI

AI Capabilities Inside Product:

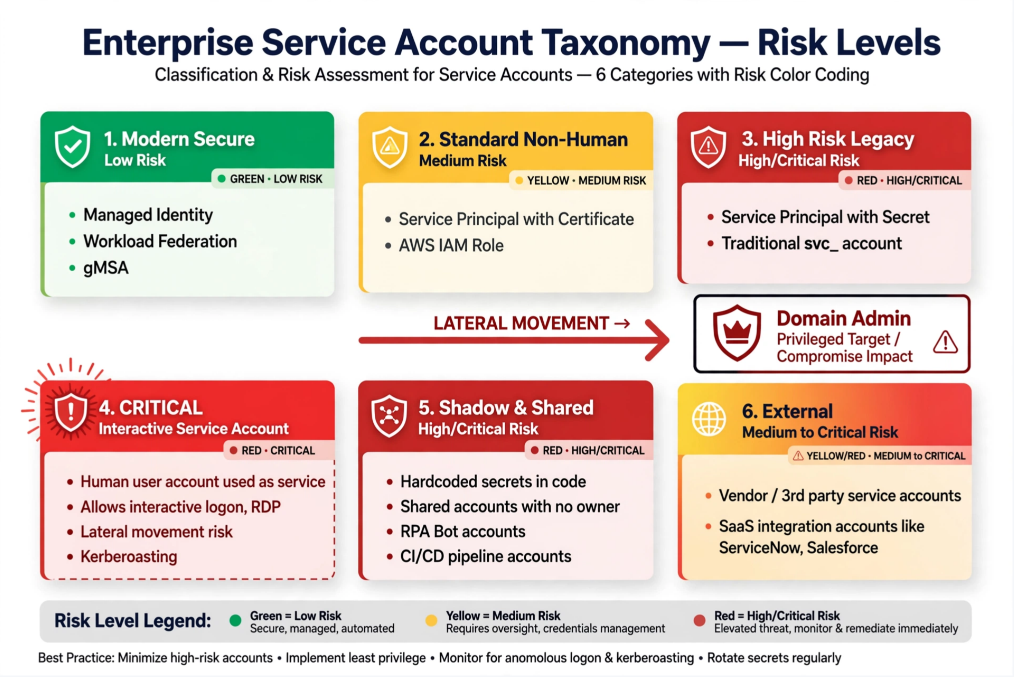
- AI Discovery: NLP scans code/GitHub/AD descriptions to find Shadow Service Accounts with no owner
- AI Risk Scoring: ML model scores Tier0/1/2 based on privilege, interactive logon, last use, data classification
- AI Owner Prediction: Predicts owner from code commit history, AD manager chain, ServiceNow ticket history (for orphaned accounts)
- AI Anomaly Detection: Sentinel + ADX + ML detects anomalous service account use (impossible travel, off-hours, ZPA bypass)
- GenAI Copilot: Converts interactive svc to gMSA/Managed Identity with auto-generated code + justification summary for auditor

By Channakeswara Vuppalamarthi | AI Product Management Portfolio | Contact via LinkedIn | Aug 2026

Problem: Critical Interactive Accounts = Lateral Movement Highway

Traditional IAM misses 50% of NHI: Interactive (Human user as service, RDP, Kerberoastable), Shadow (hardcoded secrets), RPA Bots (UiPath with mailbox access), CI/CD (GitHub Actions), SaaS Integration (ServiceNow MID)
Result: 60% secrets outside vault, <40% recertified, 14-day manual SNOW provisioning, NYDFS fine risk $2-10M

AI Product Approach:
Before: Manual CSV review to find orphaned accounts (300 hrs per audit)
After (AI): ML scans Entra SignInLogs + AD + GitHub + ServiceNow tickets -> Auto-predicts owner + risk + remediation path
Example: AI detects svc_sqlprod interactive + Domain Admin + dormant 60 days -> Auto-suggests: Convert to gMSA + generates PowerShell



AI Product Architecture: Autonomous Governance Loop

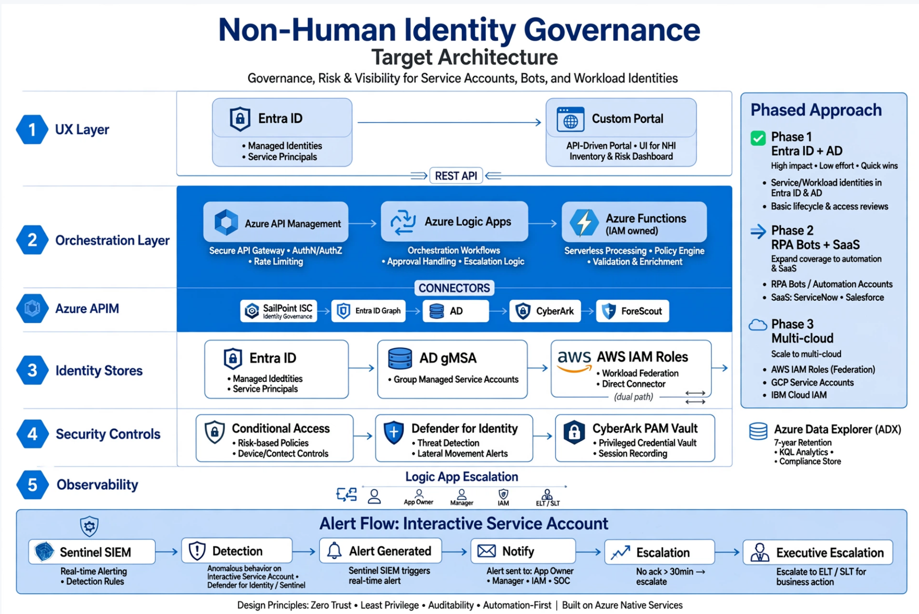
UX: ServiceNow Catalog (Single Window) -> APIM -> Logic Apps -> Functions (IAM+SOC owned)

AI Layer (New): Azure OpenAI + Azure ML + ADX KQL ML

1. Discovery Agent: NLP on code repos, AD description, SNOW tickets to find Shadow NHI
2. Risk Agent: ML classifier (features: privilege, interactive, last use, data class) -> Tier0/1/2 + configurable rotation 30-365 days
3. Owner Agent: Graph ML predicts owner from commit history + manager chain for orphaned accounts
4. Anomaly Agent: Sentinel ML + ADX detects impossible travel, ZPA bypass, off-hours use -> Auto-disable via Logic App
5. Remediation Copilot: GenAI generates gMSA/Managed Identity conversion code + business justification for ELT/SLT escalation

Controls: Conditional Access + Defender for Identity + CyberArk PAM (JIT) + ForeScout NAC + Zscaler ZPA

Observability: Sentinel Real-time + ADX 7-year retention (NYDFS) + AI-generated audit narrative



AI Product Management: Metrics, Roadmap & Business Value

Product Metrics (North Star: % NHI Autonomous Governed):

- Adoption: % service accounts auto-discovered by AI (Target 100%), % orphaned accounts auto-mapped to owner (Target 80%)
- Risk: # Interactive Tier0 500->0, Standing secrets -80%, Mean Time to Remediate (MTTR) 14 days -> 15 mins via Copilot
- AI Performance: Owner prediction accuracy >85%, Anomaly detection precision >90%, False positive <5%
- Compliance: Recertification 40%->100% via AI-nudged attestation, Evidence time 3 weeks -> 10 mins (GenAI summary)
- Business: Avoid $2-10M NYDFS fine, Save 2500 hrs/yr IAM ops, Zero P1s from expiry, -10% cyber insurance premium

Phased AI Roadmap (Less Effort High Impact First):

Phase 1 MVP (8 wks): Rule-based discovery (Entra+AD) + Risk scoring + SNOW integration + Sentinel alerts (High Impact Low Effort)

Phase 2 AI (8 wks): ML Owner Prediction + Anomaly Detection + RPA/SaaS discovery (ServiceNow, Salesforce)

Phase 3 GenAI (8 wks): Copilot for auto-remediation code + AI audit narrative + Multi-cloud AWS/GCP/IBM dual toggle (Federation vs Direct)

AI Product Management Showcase: Problem Framing -> Taxonomy -> Phased Roadmap -> AI Architecture -> Metrics

Full Charter + PRD + KQL + Logic App definitions on GitHub | By Channakeswara Vuppalamarthi | Contact via LinkedIn